

# 4th Multi-modal Aerial View Image Challenge - Translation: Our Solution

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Figure 1. Teaser: Our proposed solution for multimodal aerial-view image translation across EO, IR, and SAR modalities.

## Abstract

*In this paper, we present a comprehensive study on cross-modal remote sensing image-to-image translation.*

## 1. Introduction

The Multi-modal Aerial View Image Challenge for cross domain image translation [1–3] is a leading challenge held in conjunction with the Perception Beyond the Visible Spectrum (PBVS) workshop from 2023 to 2025. Data collected across multiple sensing modalities—such as Electro-Optical (EO), Infrared (IR), and Synthetic Aperture Radar (SAR)—each capture distinct characteristics and measurements of objects, providing complementary perspectives on the same scene. However, data coverage limitations and modality-specific constraints often hinder the full utilization of these diverse sources. For instance, while SAR sensors offer all-weather capabilities and can penetrate atmospheric obstructions, their imagery is complex to interpret and publicly available datasets are scarce. Conversely, widely available EO imagery is limited by lighting conditions and atmospheric visibility. Domain translation seeks to bridge this gap by converting one modality into another,

enabling cross-modal data synthesis and enhancing sensor interoperability. By focusing on translation tasks such as RGB→IR, SAR→EO, SAR→IR, and SAR→RGB, the MAVIC-T challenge aims to bolster existing data volume, increase data diversity across regions, and serve as a platform for advancing conditioned image synthesis techniques, ultimately fostering the development of more adaptable and robust models for multi-modal image analysis.

interpret SAR need special knowledge [4].

In this paper, we propose [YourMethodName], a novel [Diffusion/GAN/Transformer] framework designed to [solve specific problem]. Our approach incorporates a [Specific Module, e.g., 'Latent Consistency Constraint'] and a [Specific Loss] to ensure that the translated images maintain the precise geometric layout of the source modality while achieving high photorealism.

## 2. Related Work

Image translation methods in general domain across diverse sensor modalities, include grayscale, near-infrared (NIR), thermal image colorization, and color transfer functions [4–7].

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